BASIC LOOP UNDERSTANDING

1. Print Numbers From 1 to N

1. Write a Java program that **prints numbers from 1 to N using a loop**, but **skip numbers that are multiples of 3 and print the remaining numbers in a list**.

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2. Print Numbers From N to 1

2. Write a Java program that **prints numbers from N down to 1**, but **stop printing if the number becomes divisible by 7**.

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3. Print Even Numbers

3. Write a Java program to **print all even numbers between 1 and N and also return the count of even numbers**.

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4. Print Odd Numbers

4. Write a Java program that **prints all odd numbers from 1 to N and calculates their total sum**.

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5. Multiplication Table

5. Write a Java program that **prints the multiplication table of a given number up to 10 and stores the results in an array**.

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6. Numbers Divisible by 3 and 5

6. Write a Java program to **print all numbers between 1–100 that are divisible by both 3 and 5 and return their count**.

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7. Count Numbers Between 1–N

7. Write a Java program to **count how many numbers between 1 and N are divisible by 2 or 5**.

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8. Skip Multiples of 4

8. Write a Java program to **print numbers from 1 to N but skip numbers that are multiples of 4 using continue**.

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9. Print 1–100 Except 50

9. Write a Java program to **print numbers from 1 to 100 but skip 50 and also print how many numbers were printed**.

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10. Reverse Numbers Using Loop

10. Write a Java program to **print numbers from N to 1 and also store them in a reversed array**.

SUM AND ARITHMETIC LOOP PROBLEMS

11. Sum of First N Numbers

11. (Easy) Write a Java program to **find the sum of first N natural numbers**.
12. (Medium) Write a Java program to **find the sum of first N numbers using a loop and compare it with the mathematical formula result**.

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13. Sum of Even Numbers

13. (Easy) Write a Java program to **find the sum of all even numbers from 1 to N**.
14. (Medium) Write a Java program to **find the sum of even numbers and count how many even numbers exist till N**.

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15. Sum of Odd Numbers

15. (Easy) Write a Java program to **find the sum of odd numbers till N**.

16. (Medium) Write a Java program to **find both sum and average of odd numbers till N**.

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17. Sum of Digits

17. (Easy) Write a Java program to **calculate the sum of digits of a number**.
18. (Medium) Write a Java program to **calculate sum of digits repeatedly until the result becomes a single digit**.

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19. Product of Digits

19. (Easy) Write a Java program to **calculate product of digits of a number**.
20. (Medium) Write a Java program to **calculate product of digits and count how many digits were multiplied**.

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21. Average of N Numbers

21. (Easy) Write a Java program to **find the average of numbers from 1 to N**.
22. (Medium) Write a Java program to **calculate average of only numbers divisible by 3 between 1 and N**.

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23. Sum of Squares

23. (Easy) Write a Java program to **find sum of squares of first N numbers**.
24. (Medium) Write a Java program to **find sum of squares of only even numbers till N**.

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25. Sum of Cubes

25. (Easy) Write a Java program to **find sum of cubes of first N numbers**.
26. (Medium) Write a Java program to **find sum of cubes of numbers divisible by 3 till N**.

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27. Numbers Divisible by 3

27. (Easy) Write a Java program to **find sum of numbers divisible by 3 till N**.

28. (Medium) Write a Java program to **count numbers divisible by 3 and return their sum till N**.

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29. Factorial

29. (Easy) Write a Java program to **calculate factorial of a number using a loop**.
30. (Medium) Write a Java program to **calculate factorial and count how many multiplications were performed**.

DIGIT BASED LOOP PROBLEMS

31. Count Digits

31. (Easy) Write a Java program to **count number of digits in a number**.
32. (Medium) Write a Java program to **count digits and also find sum of digits**.

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33. Reverse Number

33. (Easy) Write a Java program to **reverse a number**.
34. (Medium) Write a Java program to **reverse a number and check whether original and reversed numbers are equal**.

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35. Palindrome Number

35. (Easy) Write a Java program to **check whether a number is palindrome**.
36. (Medium) Write a Java program to **check palindrome and return the reversed number as well**.

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37. Largest Digit

37. (Easy) Write a Java program to **find the largest digit in a number**.
38. (Medium) Write a Java program to **find largest digit and count how many times it appears**.

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39. Smallest Digit

39. (Easy) Write a Java program to **find the smallest digit in a number**.
40. (Medium) Write a Java program to **find smallest digit and return its position**.

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41. Reduce to Single Digit

41. (Easy) Write a Java program to **keep adding digits until a single digit remains**.
42. (Medium) Write a Java program to **count how many iterations were required to reach single digit**.

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43. Armstrong Number

43. (Easy) Write a Java program to **check whether a number is an Armstrong number**.
44. (Medium) Write a Java program to **print all Armstrong numbers between 1 and N**.

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45. Print Digits

45. (Easy) Write a Java program to **print all digits of a number**.
46. (Medium) Write a Java program to **print digits from left to right**.

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47. Swap First and Last Digit

47. (Easy) Write a Java program to **swap first and last digit of a number**.
48. (Medium) Write a Java program to **swap first and last digit and print the new number**.

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49. Check Digit 0

49. (Easy) Write a Java program to **check whether a number contains digit 0**.
50. (Medium) Write a Java program to **count how many times digit 0 appears**.

COUNTING & FREQUENCY

51. Count Even Digits

51. (Easy) Write a Java program to **count even digits in a number**.

52. (Medium) Write a Java program to **count even digits and print them**.

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53. Count Odd Digits

53. (Easy) Write a Java program to **count odd digits in a number**.

54. (Medium) Write a Java program to **count odd digits and calculate their sum**.

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55. Occurrences of Digit

55. (Easy) Write a Java program to **count occurrences of a given digit in a number**.

56. (Medium) Write a Java program to **find the digit with maximum frequency**.

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57. Numbers Divisible by 7

57. (Easy) Write a Java program to **count numbers divisible by 7 from 1 to N**.

58. (Medium) Write a Java program to **print those numbers and their count**.

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59. Count Prime Numbers

59. (Easy) Write a Java program to **count prime numbers till N**.

60. (Medium) Write a Java program to **print all prime numbers till N and count them**.

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61. Numbers Containing Digit 5

61. (Easy) Write a Java program to **count numbers containing digit 5 between 1 and N**.

62. (Medium) Write a Java program to **print those numbers containing digit 5**.

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63. Multiples of 3 and 5

63. (Easy) Write a Java program to **count numbers divisible by both 3 and 5 till N**.

64. (Medium) Write a Java program to **print those numbers and calculate their sum**.

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65. Digit Frequency

65. (Easy) Write a Java program to **count frequency of each digit in a number**.

66. (Medium) Write a Java program to **find digit with maximum frequency**.

MATHEMATICAL LOOP PROBLEMS

67. Prime Number

67. (Easy) Write a Java program to **check whether a number is prime**.

68. (Medium) Write a Java program to **check prime number using optimized loop up to √N**.

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69. Print Primes Till N

69. (Easy) Write a Java program to **print all prime numbers till N**.

70. (Medium) Write a Java program to **count prime numbers till N**.